

Implicit neural fields recover the missing wedge in small-angle scattering tensor tomography

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Abstract

Small-angle X-ray scattering tensor tomography (SASTT) resolves the orientation of nanostructure inside millimetre-sized samples, but the acquisition geometry is mechanically constrained: the rotation axis can only be tilted so far before the sample mount blocks the beam. The orientations that are never reached form a missing wedge, and because each voxel carries not a single scalar but a whole angular scattering profile—its reciprocal-space map (RSM), the intensity as a function of direction at a fixed momentum-transfer shell—that wedge removes information in the angular domain as well as in real space. Conventional voxel-wise reconstructions have no mechanism to fill it: they fit only what was measured, and the unmeasured directions are left to whatever the regulariser happens to prefer.

We approach the problem from a different direction. A SASTT volume, restricted to one momentum-transfer shell as both our method and the baselines are, is a field of spherical functions sampled from a limited set of viewing directions, and is structurally the same object that neural radiance fields were built to recover, and the sparse, limited-angle regime is exactly where implicit neural representations have proven most useful. We therefore represent the SASTT volume as a coordinate network mapping position to spherical-harmonic coefficients, and train it directly against the measured projections through a new PyTorch implementation of the `mumott` forward operator, which makes the whole pipeline differentiable and GPU-resident. No explicit smoothness prior is used. Instead the reconstruction is shaped by two implicit mechanisms: a coarse-to-fine schedule that reveals hash-grid resolutions and angular orders progressively, and a stochastic variant of that schedule that avoids a subtle failure of adaptive optimisers when parameters are held at exactly zero.

Across six datasets—three simulated phantoms, a remounted human trabecular bone measurement, a full-coverage synthetic control, and a wide-angle scattering measurement of steel wire—the method improves on both `mumott` baselines on every metric on all four datasets with exact ground truth, reducing orientation error by a factor of two to five, with the largest gains precisely in the reciprocal-space directions the missing wedge leaves unconstrained.

1 Introduction

Small-angle X-ray scattering (SAXS) reports on the size, shape and orientation of nanoscale structure averaged over the illuminated volume. Focusing the beam and raster scanning the sample turns this into an imaging method, and combining the scan with tomographic rotation extends it to three dimensions [Liebi et al., 2015, Schaff et al., 2015]. What makes the tensor variant of the technique distinctive is that the quantity reconstructed in each voxel is not a number but a function: the reciprocal-space map (RSM), describing how strongly that voxel scatters as a function of direction. It is this angular structure that carries the orientation information—the direction of mineralised collagen in bone [Georgiadis et al., 2015], of fibres in a composite—and it is why SASTT has attracted interest well beyond its original demonstrations.

The price is data. A projection is itself a two-dimensional raster scan, and projections must be recorded around two rotation axes, with the third rotation supplied by the azimuthal angle of the detector. Sampling densely enough to satisfy the Crowther criterion [Crowther et al., 1970] on both axes is out of reach at any synchrotron; measurements therefore follow empirical sampling rules and take tens of hours. But the more fundamental limitation is not density—it is reach. The rotation axis is tilted by a goniometer, and beyond roughly 45° the sample mount itself gets in the way. A whole cone of sample orientations is simply never accessible.

Nielsen et al. [2024] made the consequences of this precise. Treating the reconstruction as a three-dimensional field of functions on the unit sphere, they showed that the inaccessible tilt range is a missing wedge in the classical tomographic sense, but one that acts across real and reciprocal space at once: each direction in reciprocal space is probed over an arc of sample orientations, and for some directions that arc is empty. They also did something that makes quantitative study of the problem possible at all. By physically remounting a sample of human trabecular bone in a second orientation and merging the two acquisitions, they produced a dataset with genuinely complete angular coverage—a real measurement, not a simulation, against which limited-coverage reconstructions can be judged. Their conclusion was reassuring but bounded: means and relative anisotropies survive the missing wedge reasonably well, while sharper angular features and orientation estimates degrade.

That leaves an obvious question. The missing wedge is a property of the acquisition, and short of remounting every sample—which doubles an already expensive measurement—it cannot be removed experimentally. Can it be compensated for during reconstruction?

Conventional SASTT reconstructions treat each voxel’s coefficients as free parameters and fit them to the data, usually with a spatial smoothness penalty [Liebi et al., 2018, Nielsen et al., 2023]. Such a model has no capacity to reason about what it did not measure. In the directions the wedge removes, the data constrain nothing, and the result is whatever the regulariser prefers—typically an over-smoothed, systematically under-estimated intensity, as we show below. The missing information has to come from somewhere, and in a purely voxel-wise parameterisation there is nowhere for it to come from.

Our starting point is the observation that this is not a new kind of problem. In computer vision, neural radiance fields [Mildenhall et al., 2020] reconstruct a three-dimensional scene in which every point emits light as a function of direction, from a set of images taken along a limited set of viewing directions. Strip away the rendering specifics and the object being recovered is a field of functions on the sphere, observed through directional projections—the same mathematical structure as a SASTT volume, with the RSM shell in place of the radiance lobe. Crucially, the setting in which such representations have proven most valuable is the sparse, limited-view one, where classical per-voxel methods degrade badly and the implicit representation’s built-in bias towards spatially coherent solutions does real work [Zha et al., 2022]. That is precisely the SASTT regime.

This paper makes two contributions.

First, we recast SASTT reconstruction as the training of an implicit neural field. A coordinate network maps each position in the sample to the spherical-harmonic coefficients of that voxel’s RSM shell. Because neighbouring voxels are decoded from a shared, smoothly interpolated feature grid rather than from independent parameters, the representation ties the sample together spatially. A direction that is unconstrained at one voxel is constrained at its neighbours, and the network is obliged to find a solution consistent with both. This is where the missing-wedge robustness comes from, and it is not something a per-voxel model can express.

Second, we reimplement the `mumott` forward operator in PyTorch. The physical model is unchanged, the same projector, the same spherical-harmonic quadrature, but it is now differentiable and runs on the GPU, so gradients flow from the measured projections back to the network weights. This is what makes the first contribution possible in practice, and it is reusable: any differentiable reconstruction model can be trained through it.

We deliberately present the method *without* explicit regularisation. What shapes the reconstruction instead is a pair of implicit mechanisms described in Section 3: a coarse-to-fine schedule that grows the model’s spatial and angular capacity during training, and a stochastic version of the angular schedule that sidesteps a failure mode of adaptive optimisers we encountered when coefficients are held at exactly zero. Explicit smoothness penalties are implemented and available, and they do help on some noisy real acquisitions, but they are not needed for the results reported here and we discuss them separately in Section 5.5.

2 Related work

Tensor tomography and its reconstruction algorithms. SASTT was introduced independently by Liebi et al. [2015] and Schaff et al. [2015]. The reconstruction problem was formalised by Liebi et al. [2018], who modelled the RSM with spherical harmonics and fitted the coefficients by gradient-based optimisation, and extended by Nielsen et al. [2023], who replaced the spherical-harmonic basis with a grid of Gaussian radial basis functions to handle textures whose angular structure is too complex for a low-order harmonic expansion. Both approaches are available in the open-source `mumott` package [Nielsen et al., 2025], and both are per-voxel: the unknowns are the coefficients at each voxel, coupled only through an explicit spatial regulariser. These are the baselines we compare against.

The missing wedge. Limited-angle artefacts are a classical subject in tomography, where the unmeasured Fourier wedge produces characteristic streaking and boundary blurring; the artefacts are well characterised, and which features survive is determined by the geometry of the missing directions rather than by the reconstruction algorithm [Friel and Quinto, 2013]. In SASTT the geometry is richer, because the missing directions affect the angular as well as the spatial reconstruction. Nielsen et al. [2024] characterised this in detail and, by remounting the sample, produced the complete-coverage reference dataset we use here. Their analysis quantifies how much is lost; our aim is to recover part of it.

Implicit neural representations. Coordinate networks represent a signal as a function from position to value, trained directly against observations of that signal. Since NeRF [Mildenhall et al., 2020] they have been applied widely; the developments that matter here are the input encodings that let a small network represent high-frequency detail—Fourier features [Tancik et al., 2020], periodic activations [Sitzmann et al., 2020], and especially the multiresolution hash encoding of Müller et al. [2022], which we adopt because its per-level structure gives a natural handle on coarse-to-fine training. In tomography, Zha et al. [2022] showed that a hash-encoded neural field reconstructs cone-beam CT from far fewer projections than classical methods require, and Rückert et al. [2022] developed a related adaptive formulation. Our forward model differs fundamentally from these—the measurement is an angularly resolved scattering pattern, not an attenuation integral—but the motivation is the same: in the sparse, limited-angle regime, the representation’s inductive bias is doing the work that data alone cannot.

Frequency scheduling. Revealing high-frequency capacity gradually during training stabilises coupled, underconstrained optimisation problems. Lin et al. [2021] introduced the underlying idea, masking high-frequency positional-encoding channels early in training to stabilise joint pose-and-scene optimisation; FreeNeRF [Yang et al., 2023] showed that an analogous frequency-domain schedule substantially improves few-shot neural rendering; and Li et al. [2023] apply the same principle to a multiresolution hash encoding directly, enabling only the coarsest levels at first and progressively activating finer ones to avoid the relearning that early high-resolution instability would otherwise force—the mechanism our spatial schedule (Section 3.4) adopts. Our angular

schedule, which instead reveals spherical-harmonic degrees, has a recent counterpart in sparse-view Gaussian Splatting: Fang et al. [2026] found that stochastically dropping higher-degree spherical-harmonic coefficients, rather than truncating them outright, reduces overfitting under sparse 2D supervision. Our setting differs in that the missing directions are never observed at all rather than merely under-sampled, and in Section 3.5 we identify a distinct, optimiser-level reason that a stochastic rather than a hard curriculum is necessary there.

3 From radiance fields to reciprocal-space maps

This section gives the idea and the design decisions; the mathematical statement of the forward model and the training objective is deferred to Section 6 so that the argument can be followed without it.

3.1 What is being reconstructed

A SASTT experiment measures, for every sample orientation and every position in the raster scan, how much radiation is scattered into each azimuthal segment of the detector. The quantity we want is a field: at each voxel $\mathbf{r}$ inside the sample, a function $I(\mathbf{r}, \hat{\mathbf{q}})$ giving the scattered intensity as a function of direction $\hat{\mathbf{q}}$ on the unit sphere. Because SAXS patterns are centrosymmetric, this function is even, and it is conventionally expanded in real spherical harmonics of even degree up to a band limit $\ell_{\max}$, giving $C = (\ell_{\max}/2 + 1)^2$ coefficients per voxel ($C = 45$ for $\ell_{\max} = 8$). The $\ell = 0$ coefficient is proportional to the mean scattering intensity—essentially a conventional tomogram—and the higher orders carry the anisotropy that makes the technique worth doing.

A scoping remark that matters throughout this paper. Reciprocal space is three-dimensional: scattering intensity depends on the full momentum-transfer vector, both its magnitude q —which encodes a length scale of the sample’s ultrastructure, such as a fibril spacing or a Bragg peak—and its direction $\hat{\mathbf{q}}$, which encodes orientation. What we reconstruct is not that full three-dimensional map. It is its restriction to a single momentum-transfer shell: a fixed q , or a fixed detector-defined q -range chosen to isolate one length scale of interest, azimuthally regrouped into the direction-dependent intensity $I(\mathbf{r}, \hat{\mathbf{q}})$ introduced above. Within that shell I is a function of direction alone, and every occurrence of “reciprocal-space map” or “RSM” in this paper refers to this shell-restricted spherical function, not to the full three-dimensional reciprocal space. This is not a simplification particular to our method: it is standard practice in SASTT, and both `mumott` baselines we compare against make exactly the same restriction. A genuinely volumetric reconstruction—recovering $I(\mathbf{r}, \hat{\mathbf{q}}, q)$ jointly across shells rather than one shell at a time—is a natural extension of the representation we describe below, and we return to it in Section 7.

A measurement does not see $I(\mathbf{r}, \hat{\mathbf{q}})$ directly. Each detector segment integrates the scattering over an arc of directions determined by the sample orientation, and the detector integrates along the beam through the sample. Reconstruction is the inverse of that chain.

3.2 Why this looks like a radiance field

A neural radiance field represents a scene as a function from position to a colour that depends on viewing direction, and recovers it from images taken along a limited set of directions. The correspondence to SASTT is close enough to be useful:

Neural radiance field	SASTT
Radiance $c(\mathbf{r}, \mathbf{d})$ at a point	RSM $I(\mathbf{r}, \hat{\mathbf{q}})$ at a voxel
Camera viewing directions	Sample orientations reachable by the goniometer
Rendering integral along a ray	Beam integral through the sample
Sparse, limited set of views	Limited tilt range: the missing wedge

In both cases the object is a spatial field of directional functions, observed through a projection that mixes space and angle, from a set of directions that does not cover the sphere. The reason this correspondence is worth exploiting is not aesthetic. It is that the sparse-view regime is where neural fields earn their keep, and SASTT is permanently in that regime—not by choice, but because the goniometer cannot reach further and the beamtime is not there to sample more densely.

3.3 The representation

We represent the coefficient field with a coordinate network. A voxel position is normalised to the unit cube (isotropically, so that a hash level has the same physical resolution along every axis) and passed through a multiresolution hash encoding [Müller et al., 2022]: a stack of feature grids from coarse to fine, each storing learnable feature vectors at its corners, trilinearly interpolated at the query point and concatenated, together with the raw normalised coordinate itself. A small multilayer perceptron—two hidden layers of width 64—decodes the concatenated features into the C spherical-harmonic coefficients. The $\ell = 0$ channel passes through a softplus, since mean intensity cannot be negative, and each channel is divided by a fixed per-degree scale so that the C outputs receive gradients of comparable magnitude. Figure 2 details the encoding step itself.

The important property is what this does *not* allow. In a per-voxel model, changing one voxel’s coefficients affects nothing else; a direction unconstrained by the data at that voxel is free. Here, every voxel is decoded from feature vectors shared with its neighbours through interpolation, and through an MLP shared by the entire volume. To place an arbitrary value at one voxel, the network must move features that its neighbours also read. Spatial coherence is not added as a penalty term—it is the only thing the representation can express cheaply. This is the mechanism that lets information leak from well-measured directions and locations into poorly measured ones.

We deliberately favour grid capacity over network capacity (a wide hash encoding, a narrow MLP). An oversized shared trunk tends to compress genuinely different per-voxel features into a few directions it finds convenient, which shows up as the same RSM shape appearing at every voxel, merely rescaled.

3.4 Growing the model during training

Training the full model from a random start does not work well: with all hash levels and all harmonic degrees active from the first iteration, the optimiser fits high-frequency structure to noise long before the coarse structure is right. Restricting capacity early to stabilise this kind of coupled, underconstrained optimisation is the coarse-to-fine principle discussed in Section 2 [Lin et al., 2021, Yang et al., 2023]; we grow the model along two axes.

Spatially, the hash levels are revealed from coarse to fine over the first part of training, with a soft edge on the level currently appearing. Early on the network can only express smooth, large-scale structure; fine detail becomes available once that structure is settled. This is the same mechanism Li et al. [2023] use to avoid the relearning that early high-resolution instability would otherwise force, and the tomographic analogue of the frequency schedules discussed in Section 2.

Angularly, the harmonic degrees are revealed in order, $\ell = 0$ first, then $\ell = 2, 4, \dots$. The reconstruction therefore starts as an isotropic tomogram—a well-conditioned problem—and anisotropy is introduced only once that is in place. Since the higher degrees are exactly what the missing

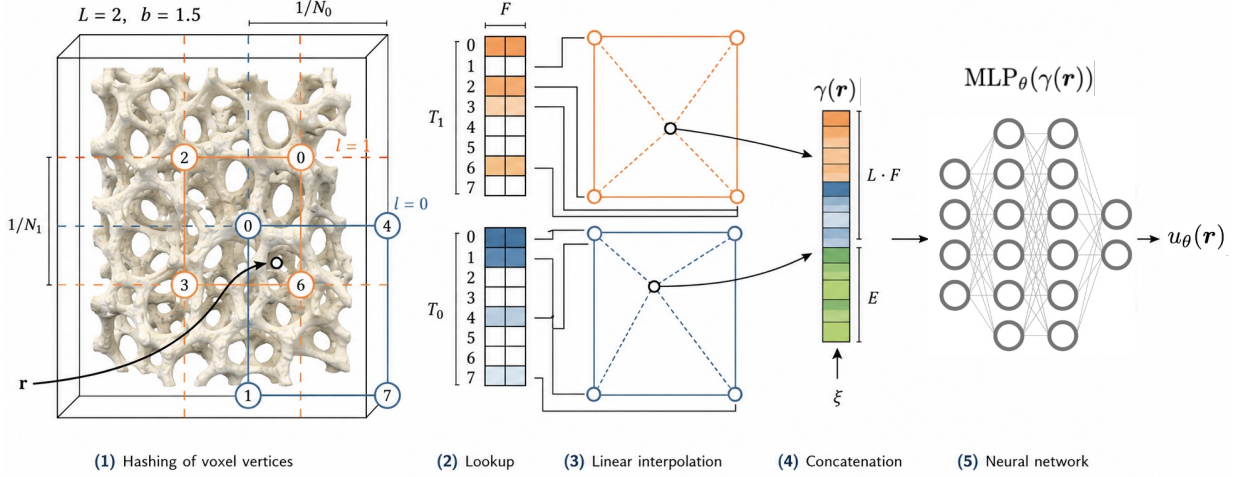


Figure 1: Schematic of the multiresolution hash encoding [Müller et al., 2022] that produces the coordinate embedding $\gamma(\mathbf{r})$ entering the representation $u_\theta(\mathbf{r}) = s \odot \sigma(W \text{MLP}_\theta(\gamma(\mathbf{r})))$ of Section 6. (1) For an input position $\mathbf{r}$ we find the surrounding voxel at each of the L resolution levels and hash the integer coordinates of its corners to obtain per-corner table indices. (2) For every corner index we look up the corresponding F -dimensional feature vector from that level’s hash table, and (3) linearly interpolate the corner features according to the position of $\mathbf{r}$ within the level- j voxel. (4) The interpolated features from all L levels are concatenated with the raw coordinate itself (the auxiliary input $\xi = \mathbf{r}$ shown in the figure) to form $\gamma(\mathbf{r})$, which (5) is passed through MLP_θ and the linear readout W to predict the spherical-harmonic coefficients. During training, gradients are backpropagated through the MLP (5), the concatenation (4) and the interpolation (3), and accumulate in the looked-up corner features—the mechanism, discussed in Section 3, that ties neighbouring voxels together and lets well-measured directions constrain poorly measured ones.

wedge leaves ambiguous, it matters that they are fitted last, on top of a solution that is already correct at low order.

Together these replace the role usually played by an explicit prior. Nothing in the objective penalises roughness; instead, roughness is simply not representable until late in training, by which time the coarse solution constrains where it can go.

3.5 A subtlety: zeros and adaptive optimisers

Implementing the angular schedule in the obvious way—masking not-yet-revealed coefficients to zero—introduces a problem that took us some time to identify.

While a degree is masked, the corresponding rows of the output layer receive exactly zero gradient. They do not move, which is intended. But the output layer is a single parameter tensor, and Adam [Kingma and Ba, 2015] maintains one step counter per tensor, incremented every iteration regardless of whether a particular row received a gradient. Bias correction is computed from that counter. By the time a degree is unlocked, its rows are treated as though they had been training all along: they receive none of the effective step-size boost that genuinely fresh parameters get early in training, and they start from a stale second-moment estimate. The higher harmonic degrees—the ones carrying the orientation signal—are precisely the ones penalised.

The standard engineering answer to a parameter that does not always receive a gradient is PyTorch’s `SparseAdam`, but it requires genuinely sparse, index-based gradients rather than a densely computed output that is masked to zero afterwards, so it does not apply here. Rather than patch the optimiser state directly, we remove the exact zeros instead. In the stochastic variant of the schedule, each voxel independently draws a coin at each iteration and, with probability

equal to the schedule’s current progress, sees *all* degrees instead of the truncated set. Early in training few voxels do, so the reveal is still gradual in aggregate; but no parameter is ever frozen at exactly zero, and every degree receives genuine, full-strength gradient from the first iteration. The coarse-to-fine behaviour is preserved statistically while the pathology is removed at its source.

We found this to be the difference between a reconstruction that resolves per-voxel RSM shape and one that collapses to a single shared shape rescaled across the volume.

A similar stochastic mechanism was independently found beneficial for a related curriculum, in a different representation and for a different stated reason: Fang et al. [2026] anneal spherical-harmonic degrees in sparse-view Gaussian Splatting by having each primitive draw a fresh truncated-or-full decision every iteration (Section 2), and attribute the improvement over hard truncation entirely to reduced overfitting under sparse 2D supervision. The mechanism identified above—stale Adam moment estimates on rows frozen at exactly zero—gives a second, distinct explanation for why a stochastic curriculum can outperform a hard one, complementary to theirs rather than competing with it.

3.6 Two-phase training

The schedules above define a first phase that reliably produces a clean, spatially coherent, low-order-correct solution, but a conservative one: by construction it spends most of its budget at limited capacity and a low learning rate. A second phase warm-starts from it with a fresh optimiser, a substantially higher learning rate, and every constraint lifted—full spatial and angular capacity from its first iteration. Phase one establishes where the sample is and roughly how it scatters; phase two sharpens the per-voxel angular detail on top of that.

Splitting training this way matters because the two stages want incompatible learning rates. Running phase one longer does not substitute for phase two, and starting at phase two’s learning rate reproduces the failure phase one exists to prevent.

3.7 The forward operator in PyTorch

Training requires gradients of the measured signal with respect to the coefficient field, which means the forward model must be differentiable. We reimplemented `mumott`’s projector and spherical-harmonic quadrature in PyTorch [Paszke et al., 2019], preserving the physics—including the Ewald-sphere curvature correction needed for wide-angle data [Carlsen et al., 2024], where the detector segments no longer lie in a plane through the origin. The reimplementation is validated against `mumott` on the same geometries.

Beyond enabling this work, making the operator differentiable and GPU-resident is useful in its own right: it turns SASTT reconstruction into a standard optimisation problem in a mainstream autodiff framework, so any differentiable model or learned prior can be trained through the same physics.

4 Datasets

We evaluate on six datasets chosen to separate the effects we care about: exact ground truth versus real measurement, missing wedge versus complete coverage, and small- versus wide-angle scattering. Table 1 summarises them.

Simulated phantoms with exact ground truth. The three `nielsen` phantoms [Nielsen et al., 2023] differ in the symmetry of their reciprocal-space maps: **M** is zonal (rotationally symmetric about an axis, harmonics to $\ell = 12$), **T** is fibre-like (pure symmetric rank-2 tensors, $\ell \leq 2$), and **mammoth** has unrestricted symmetry ($\ell \leq 8$). Each is simulated with the standard acquisition geometry: full rotation about the beam-orthogonal axis, but tilt limited to 45° , giving

Table 1: Datasets. “Wedge” is the tilt limit of the acquisition; 45° is the standard geometric constraint, “none” means full angular coverage. $\ell_{\max}$ is the band limit used by all methods on that dataset.

Dataset	Type	Ground truth	Wedge	$\ell_{\max}$	Volume
nielsen-m	simulated, zonal RSM	exact	45°	12	50^3
nielsen-t	simulated, rank-2 RSM	exact	45°	2	50^3
nielsen-mammoth	simulated, general RSM	exact	45°	8	$60 \times 60 \times 80$
fiber-synthetic-full	simulated fibres	exact	none	8	100^3
trabecular-bone	human trabecular bone (SAXS)	remounted	45°	8	$65 \times 66 \times 65$
steel-wire-waxs	steel wire (WAXS)	none	45°	8	60^3

417 projections over nine tilt levels with eight azimuthal detector segments. Because the exact coefficient field is stored with the data, these give an unambiguous reference. We use the lowest-noise realisation of each.

A full-coverage control. `fiber-synthetic-full` is constructed the same way as the phantoms—coefficient vectors assigned voxel-wise to a composite fibre volume, then forward-projected—but its projection directions span the whole orientation space with no missing wedge. It isolates what the method does when nothing is missing, separating general reconstruction quality from missing-wedge compensation.

Real data, evaluated without a circular reference. The `trabecular-bone` dataset is the remounted human trabecular bone measurement of Nielsen et al. [2024]. Two acquisitions of the same sample at different mount orientations (`main` and `remount`) each individually suffer the missing wedge; their union has near-complete coverage. The usual way to exploit this is to reconstruct the combined data and treat that as ground truth for scoring single-mount reconstructions. We deliberately do not do this. A combined-data reconstruction still has to be produced by some algorithm, and on a general, complex texture like trabecular bone that algorithm is itself one of `mumott`’s two methods (Gaussian radial basis functions, Nielsen et al., 2023). Scoring reconstruction methods against a reference produced by one of them is not a comparison against an independent truth: it structurally rewards whatever method shares that reference’s inductive bias, which is exactly the comparison we are trying to make fairly. Unlike the phantoms in Table 1, whose ground truth is exact by construction and involves no reconstruction algorithm at all, `trabecular-bone` has no such independent reference.

What it does have is something arguably more valuable: a cross-mount test that requires no reconstructed reference of any kind. Reconstruct from one mount, then predict the projections of the other. Those projections come from a physically independent acquisition at orientations the model never saw, so this checks generalisation directly in the measured-data domain rather than through the lens of a possibly biased reconstruction. We report it as the primary quantitative result for this dataset, and show the qualitative figures below against the combined reconstruction purely for visual orientation, labelled accordingly rather than as ground truth.

Wide-angle scattering. `steel-wire-waxs` is a wide-angle (WAXS) tensor-tomographic scan of a tangled knot of hardened steel wire [Carlsen et al., 2024], azimuthally regrouped around an austenite Bragg peak. It differs from the others in two ways: the detector segments have a nonzero scattering angle, so Ewald-sphere curvature must be handled in the forward model, and there is no ground truth of any kind. Reconstructions are assessed by holding out 15 % of projections and measuring how well each method predicts them.

5 Results

All methods reconstruct at the same band limit on a given dataset and see identical training data. Baselines are the two **mumott** algorithms: **mumott SH** (spherical harmonics, Liebi et al., 2018) and **mumott GK** (Gaussian radial basis functions, Nielsen et al., 2023), both run for 20 L-BFGS iterations with Laplacian weight 0.1. Metric definitions are given in Appendix A; in brief, RSM correlation measures the recovery of angular *shape*, relative-anisotropy error the recovery of anisotropy magnitude, orientation error the recovery of the principal axis, and PSNR/SSIM/NRMSE the quality of the isotropic ($\ell = 0$) volume.

5.1 Qualitative: what the missing wedge does

The clearest way to see the missing wedge is to look at a single reciprocal-space direction. Contracting the coefficient field with the spherical-harmonic basis evaluated at one unit vector $\hat{q}$ collapses it to a scalar volume $I_{\hat{q}}(\mathbf{r})$: the spatially resolved intensity scattered along that direction. Different $\hat{q}$ are constrained by the data to very different degrees. With a 45° tilt limit, some directions are probed over a full arc of sample orientations, while others—those lying near the plane that the goniometer cannot bring into the beam—are never probed at all.

Figure 3 shows the worst case on the real bone data: a direction whose missing arc is a full 180° . The left-most column is the combined two-mount acquisition reconstructed with **mumott GK**—not an independent ground truth, for the reason discussed in Section 4, but a useful visual anchor since it sees nearly the whole sphere. It shows sharp trabecular struts. Both single-mount **mumott** reconstructions blur them substantially and, as the residual panels make clear, systematically *under-estimate* the intensity throughout the structure—the characteristic signature of a fit that has no information along this direction and falls back on its smoothness prior. Tellingly, this includes the single-mount **mumott GK** reconstruction, even though it shares an algorithm with the reference column: matching inductive bias does not rescue a method from a direction the data simply do not constrain. The neural field recovers the struts with far smaller and much less structured residuals.

Figure 4 makes the same point on a simulated phantom where the ground truth is exact rather than reconstructed, ruling out any concern that the reference itself is biased. The pattern is identical and, if anything, starker: the baselines lose the object’s internal contrast and leak intensity beyond its boundary, while the neural field’s residual is close to structureless.

For contrast, Figure 5 shows a direction on the same bone dataset that the acquisition covers fully. Here the three methods are much closer together, as they should be: with complete angular information the reconstruction problem is well posed and the choice of representation matters far less. The gap in Figures 3–4 is therefore attributable to the missing wedge specifically, not to a general difference in fitting quality.

5.2 Quantitative: datasets with ground truth

Table 2 reports the four datasets with exact ground truth. The neural field wins essentially everywhere. On **nielsen-t** and **nielsen-mammoth** the margins are large across every metric; on **nielsen-m** it leads on all six; on the full-coverage control it leads on the angular metrics while **mumott GK** is marginally ahead on two isotropic-volume metrics.

Two aspects deserve comment. The orientation errors—the quantity a SASTT experiment exists to measure—improve by more than a factor of two against the better of the two baselines on every one of these datasets, ranging from $2.1\times$ on **nielsen-m** (4.5° versus 9.4°) to $5.5\times$ on **fiber-synthetic-full**. And on the full-coverage control the gap narrows markedly, consistent with the qualitative picture: when nothing is missing, representation matters less.

Table 2: Datasets with ground truth. All methods trained on the full dataset with no holdout. Best value per column in bold. Arrows indicate the direction of improvement.

Dataset	Method	RSM corr. $\uparrow$	RA MAE $\downarrow$	Orient. err. $\downarrow$	PSNR $\uparrow$	SSIM $\uparrow$	NRMSE $\downarrow$
nielsen-m	NAF (ours)	0.921	0.045	4.53°	8.29	0.658	3.592
	mumott SH	0.839	0.102	17.97 $^\circ$	-3.24	0.314	13.549
	mumott GK	0.863	0.050	9.40 $^\circ$	1.28	0.351	8.057
nielsen-t	NAF (ours)	0.990	0.007	9.51°	25.67	0.969	0.214
	mumott SH	0.782	0.052	34.19 $^\circ$	7.13	0.382	1.810
	mumott GK	0.715	0.067	36.12 $^\circ$	5.64	0.358	2.150
nielsen-mammoth	NAF (ours)	0.948	0.020	12.46°	21.65	0.852	0.577
	mumott SH	0.674	0.028	39.09 $^\circ$	4.19	0.272	4.302
	mumott GK	0.776	0.050	33.94 $^\circ$	7.30	0.349	3.008
fiber-synthetic-full (no missing wedge)	NAF (ours)	0.986	0.040	2.30°	19.04	0.895	0.276
	mumott SH	0.938	0.084	12.56 $^\circ$	3.59	0.445	1.634
	mumott GK	0.896	0.193	15.77 $^\circ$	5.92	0.518	1.249

Table 3: Cross-mount generalisation on **trabecular-bone**: NRMSE of predicted versus measured projections for the mount *not* used in training. No reconstructed reference is involved on either side of this comparison, so it is the one quantitative result on this dataset that cannot be biased by which algorithm produced a “ground truth.”

Method	main $\rightarrow$ remount	remount $\rightarrow$ main
NAF, no penalty	0.192	0.196
NAF, +penalty	0.171	0.181
mumott SH	0.315	0.332
mumott GK	0.214	0.213

5.3 Quantitative: real data via cross-mount consistency

For **trabecular-bone** we report only the cross-mount test of Table 3, for the reason given in Section 4: the alternative—reconstructing the combined data and scoring single-mount reconstructions against it—uses a reference that is itself produced by **mumott GK**, which makes any comparison involving that method circular. The cross-mount test avoids this entirely. A reconstruction from one mount is projected through the *other* mount’s real geometry and compared to its actual measurements; no reconstructed reference of any kind enters, and the test orientations were physically inaccessible during training.

On this test the neural field wins outright. Both settings beat both baselines in both directions. Against **mumott GK**, the closest competitor, the unregularised model reduces the error by 10 % (**main $\rightarrow$ remount**) and 8 % (**remount $\rightarrow$ main**); with the smoothness penalty enabled the reductions grow to 20 % and 15 %. Against **mumott SH** the margin is 39 %–46 % throughout. Generalising to unseen orientations is exactly the capability the missing-wedge problem demands, and this is the one dataset in the study where that capability is tested directly, without passing through any reconstruction that could bias the comparison.

5.4 Wide-angle scattering without ground truth

On **steel-wire-wax**s no reference of any kind exists, so we hold out 15 % of projections and measure prediction error on them (Table 4). The neural field is best, though the margin over **mumott SH** is modest. This is the hardest dataset in the study for every method—the absolute errors are about three times larger than on the bone data—and it is the only wide-angle one, requiring the Ewald-curvature-corrected forward operator. We report it because a method that

Table 4: steel-wire-waxs: held-out projection NRMSE (15 % of projections withheld). No ground truth exists for this sample.

Method	Held-out NRMSE↓
NAF (ours)	0.556
mumott SH	0.574
mumott GK	0.612

only worked on small-angle data would be much less useful, not because it is where the approach shines.

5.5 On explicit regularisation

The results above use no explicit smoothness penalty. The implementation does provide angular ($\ell(\ell + 1)$) and spatial total-variation penalties, with weights calibrated automatically against the data-loss magnitude measured at the end of the first training phase—necessary because the absolute scale of the coefficient field varies by more than two orders of magnitude across datasets, so a fixed weight is meaningless.

We tested both settings on every dataset. The effect is not uniform. Regularisation helps clearly on the real bone data—it widens the cross-mount lead over **mumott** GK from 10 %/8 % to 20 %/15 % (Table 3). It is neutral on **nielsen-m** and **nielsen-t**. And it hurts on **nielsen-mammoth**, **fiber-synthetic-full** and **steel-wire-waxs**: on the full-coverage synthetic control it costs more than 13 dB of PSNR, and on the wide-angle data it moves the method from first to third place. The pattern is consistent with regularisation helping where there is genuine measurement noise to suppress and hurting where the fit is already noise-free—the synthetic datasets have an exact forward model and nothing to guard against.

We therefore present the unregularised method as the default, and note that on noisier real acquisitions the penalty is worth enabling. Choosing it per dataset would improve several numbers above; we have not done so, preferring one configuration everywhere.

6 Mathematical formulation

We now state the model that the preceding sections described.

6.1 Forward model

Let $u : \Omega \rightarrow \mathbb{R}^C$ denote the coefficient field over the sample volume Ω , so that the reciprocal-space map at voxel $\mathbf{r}$ is

$$I(\mathbf{r}, \hat{\mathbf{q}}) = \sum_{c=1}^C u_c(\mathbf{r}) Y_c(\hat{\mathbf{q}}), \quad (1)$$

where $\{Y_c\}$ are the real spherical harmonics of even degree $\ell \leq \ell_{\max}$ ordered as in **mumott**, and $C = (\ell_{\max}/2 + 1)^2$.

A measurement is indexed by a projection n (a sample orientation), a detector pixel p within the raster scan, and an azimuthal detector segment m . The sample orientation is a rotation $R_n \in SO(3)$. Two operations produce the predicted signal. First, the tomographic projector P_n integrates the coefficient field along the beam:

$$(P_n u)_p = \int_{\text{ray}(n,p)} u(\mathbf{r}) \, ds \in \mathbb{R}^C. \quad (2)$$

Because the harmonic expansion is linear in u , this can be applied coefficient-wise. Second, each detector segment integrates the resulting RSM over the arc $\mathcal{A}_{n,m} \subset S^2$ of reciprocal-space

directions it subtends:

$$g_{n,p,m}(u) = \frac{1}{|\mathcal{A}_{n,m}|} \int_{\mathcal{A}_{n,m}} \sum_c (P_n u)_{p,c} Y_c(R_n^\top \hat{\mathbf{q}}) d\hat{\mathbf{q}} = \sum_c B_{n,m,c} (P_n u)_{p,c}, \quad (3)$$

where $B_{n,m,c}$ collects the precomputed quadrature of the c -th harmonic over segment m at orientation n . For small-angle scattering the arc lies in the plane orthogonal to the beam; for wide-angle data the finite scattering angle 2θ tilts it off that plane, and B must be computed on the corresponding small circle of the Ewald sphere. Both cases are handled.

The full forward operator $\mathcal{F} : u \mapsto g$ is linear in u and is exactly the model implemented in `mumott`; our contribution is a differentiable, GPU-resident implementation of it.

6.2 The missing wedge

The goniometer restricts the reachable orientations to those with tilt at most α (typically 45°). For a fixed reciprocal-space direction $\hat{\mathbf{q}}$, the set of orientations at which some detector segment’s arc passes through $\hat{\mathbf{q}}$ is a subset of the circle; the complement is the *missing arc* $\mu(\hat{\mathbf{q}})$. Directions with $\mu(\hat{\mathbf{q}}) = 0$ are fully measured; directions with $\mu(\hat{\mathbf{q}})$ near 180° are never measured, and the scalar volume $I_{\hat{\mathbf{q}}}$ is determined only through the coupling between harmonic coefficients. Following Nielsen et al. [2024], we compute μ analytically from α and the goniometer axis, and use it to select the directions shown in Section 5.

6.3 Representation and objective

The coefficient field is parameterised as

$$u_\theta(\mathbf{r}) = s \odot \sigma(W \text{MLP}_\theta(\gamma(\mathbf{r}))), \quad (4)$$

where γ is the multiresolution hash encoding [Müller et al., 2022], W the linear readout, s a fixed per-degree scale $1/(\ell + 1)$, and σ the identity on all channels except $\ell = 0$, where it is a softplus enforcing non-negative mean intensity.

Writing y for the measured data and w for the per-pixel validity weights, training minimises

$$\mathcal{L}(\theta) = \sum_{n \in \mathcal{B}} \|w_n \odot (\mathcal{F}_n(u_\theta) - y_n)\|_\delta, \quad (5)$$

where $\|\cdot\|_\delta$ is the Huber norm and $\mathcal{B}$ a minibatch of projections. No penalty on u_θ appears in (5): the regularisation discussed in Section 5.5 would add terms $\lambda_{\text{sh}} \sum_c \ell_c(\ell_c + 1) u_c^2$ and $\lambda_{\text{tv}} \|\nabla u\|^2$, both averaged rather than summed over voxels, and both are set to zero for the results reported here.

6.4 Schedules

Let $p = t/T$ be normalised training progress. The spatial schedule assigns level j of the hash encoding a weight

$$\omega_j(p) = \text{clip}\left(\frac{p}{f_s}(L - 1) - (j - 1), 0, 1\right), \quad \omega_0 \equiv 1, \quad (6)$$

so levels fade in from coarse to fine over the first fraction f_s of training, with a soft edge on the level currently appearing. The angular schedule defines an analogous per-degree weight $\eta_\ell(p)$ over the first fraction f_a .

In the deterministic form, coefficient c is multiplied by $\eta_{\ell_c}(p)$. In the stochastic form of Section 3.5, each voxel $\mathbf{r}$ draws $b(\mathbf{r}) \sim \text{Bernoulli}(\min(p/f_a, 1))$ independently at each step, and the mask applied at that voxel is

$$m_c(\mathbf{r}) = \begin{cases} 1 & \text{if } b(\mathbf{r}) = 1, \\ \eta_{\ell_c}(p) & \text{otherwise.} \end{cases} \quad (7)$$

In expectation this interpolates between the truncated and full models exactly as the deterministic schedule does, but no coefficient is identically zero at any step, which is what removes the optimiser pathology.

6.5 Training configuration

Phase one runs 2001 iterations at learning rate 2×10^{-4} with $f_s = 0.5$, $f_a = 0.6$, the stochastic angular schedule, and the readout W initialised to zero. Phase two warm-starts from it and runs 1500 iterations at 5×10^{-3} with both schedules disabled. Both phases use Adam [Kingma and Ba, 2015] with a $10\times$ higher learning rate on the hash grid than on the MLP, a batch of 100 projections, and the Huber loss with $\delta = 1$. The hash encoding uses 8 levels with 8 features each; the MLP has two hidden layers of width 64.

7 Conclusions

The missing wedge in SASTT is a property of the instrument, not of the sample, and it cannot be removed without doubling the measurement. What it can be is compensated for, and the results here suggest that the choice of representation is what determines how well.

Recasting the reconstruction as an implicit neural field—motivated by the structural similarity between a SASTT volume and a radiance field, and by the fact that both are recovered from a limited set of directions—improves on the established `mumott` baselines on every metric across all four datasets with exact ground truth. The gains are concentrated where the theory says they should be: in the reciprocal-space directions the acquisition never probes, where per-voxel methods have nothing to fall back on but their smoothness prior and consequently blur and under-estimate. Orientation error—the quantity these experiments exist to measure—improves by at least a factor of two on every one of them.

On real data the same result holds, but only once the comparison is set up so that it cannot be won by construction. The natural reference for the bone dataset—a reconstruction of the combined, near-complete data—is itself produced by one of the methods under test, which makes scoring against it an unreliable measure of relative accuracy. Falling back to the one protocol that sidesteps this, predicting a physically independent remounted acquisition’s projections from a model that never saw them, the neural field is the clear winner, by 8 % to 20 % over the closer baseline depending on configuration and around 40 % over the other. We regard this as a more trustworthy comparison than any metric computed against a reconstructed reference could be, and it is the one we would recommend for any future dataset that, like this one, lacks a ground truth that is exact by construction.

Two aspects of the method are worth emphasising because they are easy to overlook. The first is that no explicit prior is doing this work. The coarse-to-fine schedules and the shared feature grid shape the solution implicitly, and adding smoothness penalties helps only on the noisiest real data. The second is the stochastic angular reveal: what looked like an implementation detail turned out to be the difference between resolving per-voxel angular structure and collapsing to a single shape, because masking coefficients to exactly zero interacts badly with per-tensor step counters in adaptive optimisers. We suspect this affects other scheduled-capacity training schemes and is worth checking for.

The differentiable PyTorch forward operator is a contribution in its own right. It puts SASTT reconstruction inside a standard autodiff framework, so learned priors, uncertainty estimation and acquisition-time feedback all become straightforward to build—the last of which is what originally motivated this work, since a reconstruction available while the sample is still on the goniometer can inform where to measure next.

Everything above is deliberately restricted to a single momentum-transfer shell, as flagged in Section 3 and as is standard practice in the field. That restriction is not fundamental to the

representation. Because the network already maps a coordinate to an output, conditioning it jointly on shell as well as position is a small change—an extra input dimension rather than a new architecture—and it lets shells share the spatial feature grid instead of being reconstructed independently, which is what every method in this paper does. In preliminary experiments outside the scope of this paper we trained such a joint model across several shells simultaneously and found that shells deliberately excluded from training were reconstructed nearly as well as shells that were included, suggesting that the coupling between shells is a further source of the implicit prior that the spatial coupling already provides within one shell. Turning this into a fully volumetric—directional *and* radial—reciprocal-space reconstruction, and establishing whether it helps with the missing wedge specifically rather than just with data efficiency generally, is the extension of this work we think is most worth pursuing.

The obvious limitation is that we evaluate one architecture and one training recipe. We have not explored how far the band limit can be pushed, how the method behaves at very low projection counts, or whether the representation can be shared across samples. Given that reconstruction quality in the missing wedge is set by the representation’s inductive bias rather than by the data, those seem the questions most likely to matter.

Data and code availability

The reconstruction code, the exact commands reproducing every number in Tables 2–4, and the figure-generation scripts are available in the project repository. The bone dataset is from Nielsen et al. [2024]; the simulated phantoms are from Nielsen et al. [2023].

A Metrics

All voxel-level metrics are evaluated inside a sample mask and at a fixed set of $K = 30$ quasi-uniform reciprocal-space directions, so that no method is favoured by directions it happens to fit well.

RSM correlation is the Pearson correlation, per voxel, between the ground-truth and predicted intensities across the K directions, averaged over the mask. It measures the recovery of angular *shape* independently of absolute scale, which is the physically meaningful quantity: orientation and anisotropy depend on the shape of the RSM, not its normalisation.

Relative anisotropy (RA) is the coefficient of variation of the intensity across directions, $\text{std}_k(I)/\text{mean}_k(I)$: zero for an isotropic voxel, larger for strongly oriented ones. We report its mean absolute error against the ground truth.

Orientation error is the angle between the predicted and ground-truth principal axes, obtained from `mumott`’s approximate-symmetry-axis estimate, averaged with weights proportional to RA. The weighting matters: in a nearly isotropic voxel the principal axis is not defined and its estimate is noise.

PSNR, SSIM [Wang et al., 2004] **and NRMSE** are computed on the $\ell = 0$ channel only—the isotropic scattering volume, which is the closest analogue of a conventional tomogram. They capture spatial fidelity but are blind to angular structure, so they should be read alongside the metrics above rather than on their own.

Held-out and cross-mount NRMSE require no ground truth. The reconstruction is pushed back through the forward operator onto projections excluded from training, and compared with the actual measurements. In the cross-mount case those projections come from a separate physical acquisition of the same sample at orientations that were geometrically inaccessible during training.

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trabecular-bone / main — RSM direction $k=0$ $q=(1.00, 0.02, -0.00)$, missing arc 180°

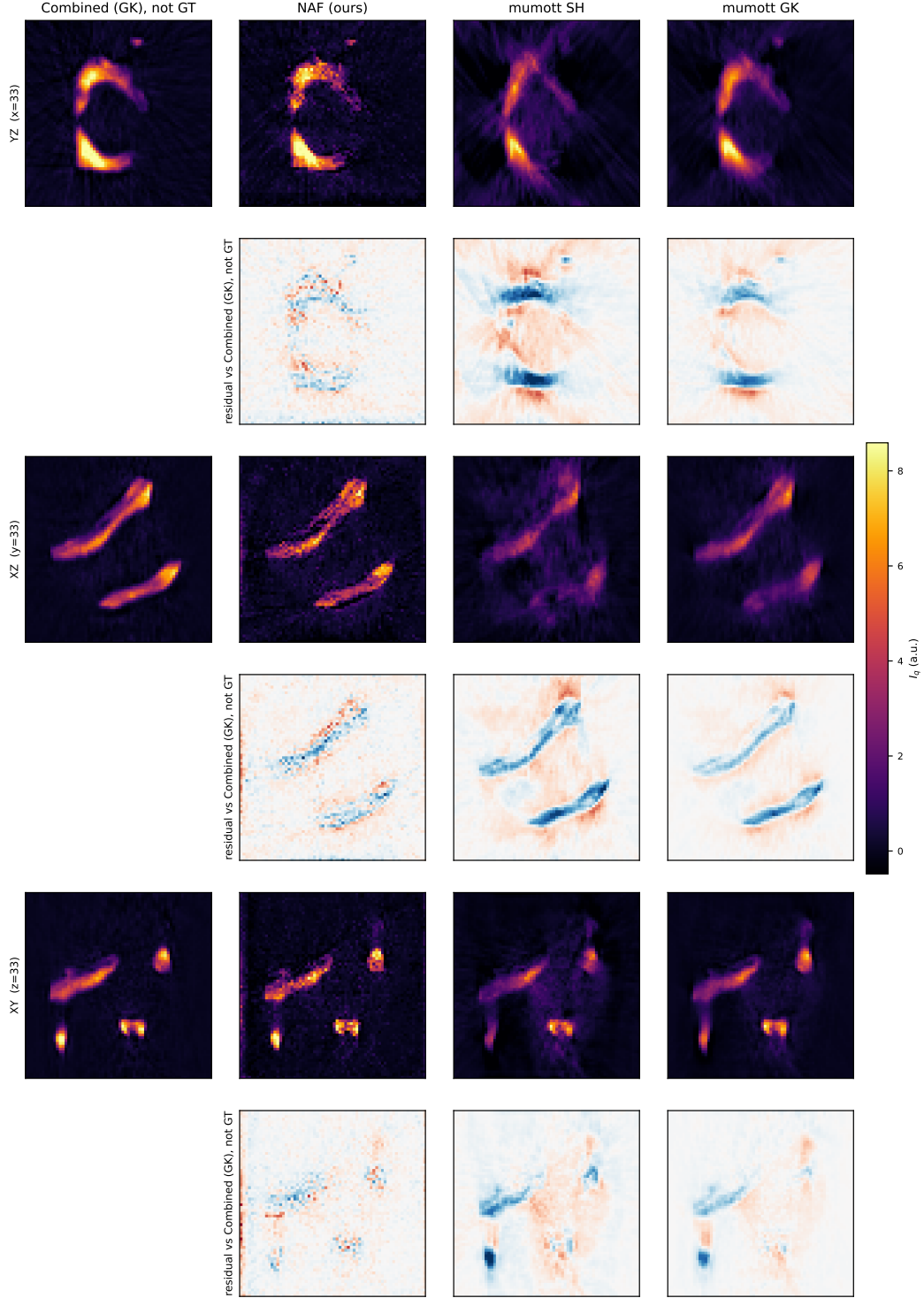


Figure 2: Human trabecular bone (`trabecular-bone`, `main` mount) along the worst-case reciprocal-space direction, whose missing arc spans 180° . Rows alternate between the reconstructed scalar volume I_q on three orthogonal planes and the residual against the left-most column, a `mumott` GK reconstruction of the combined two-mount data shown for visual orientation only (see Section 4 for why we do not treat it as ground truth). Both single-mount baselines blur the trabecular structure and under-estimate it systematically (blue); the neural field does not.

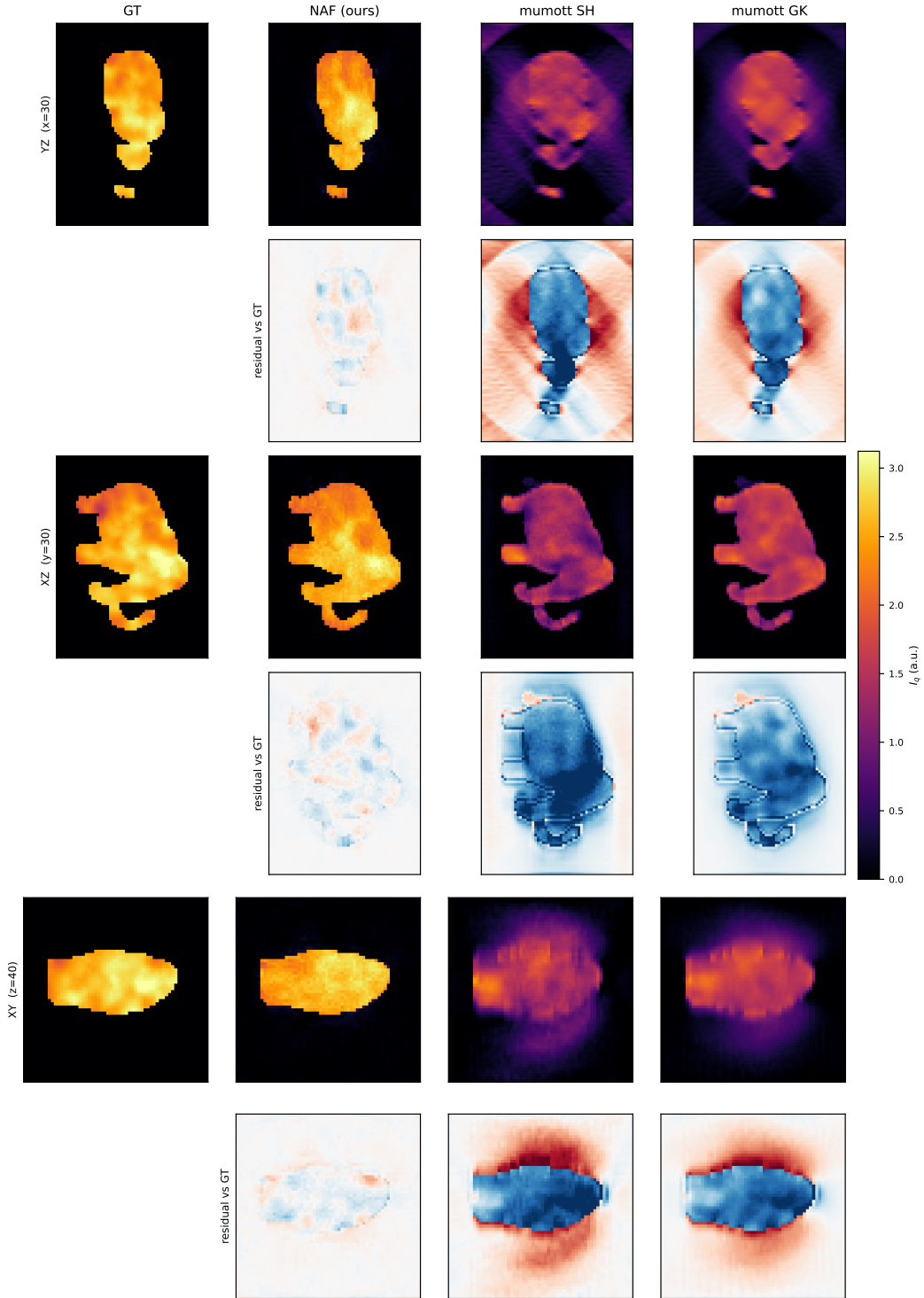


Figure 3: The nielsen-mammoth phantom along the worst-case direction, against the *exact* ground truth. The baselines under-estimate the interior and blur past the object boundary; the neural field’s residual is nearly featureless.

trabecular-bone / main — RSM direction $k=2$ $q=(0.09, 0.08, -0.99)$, missing arc 0°

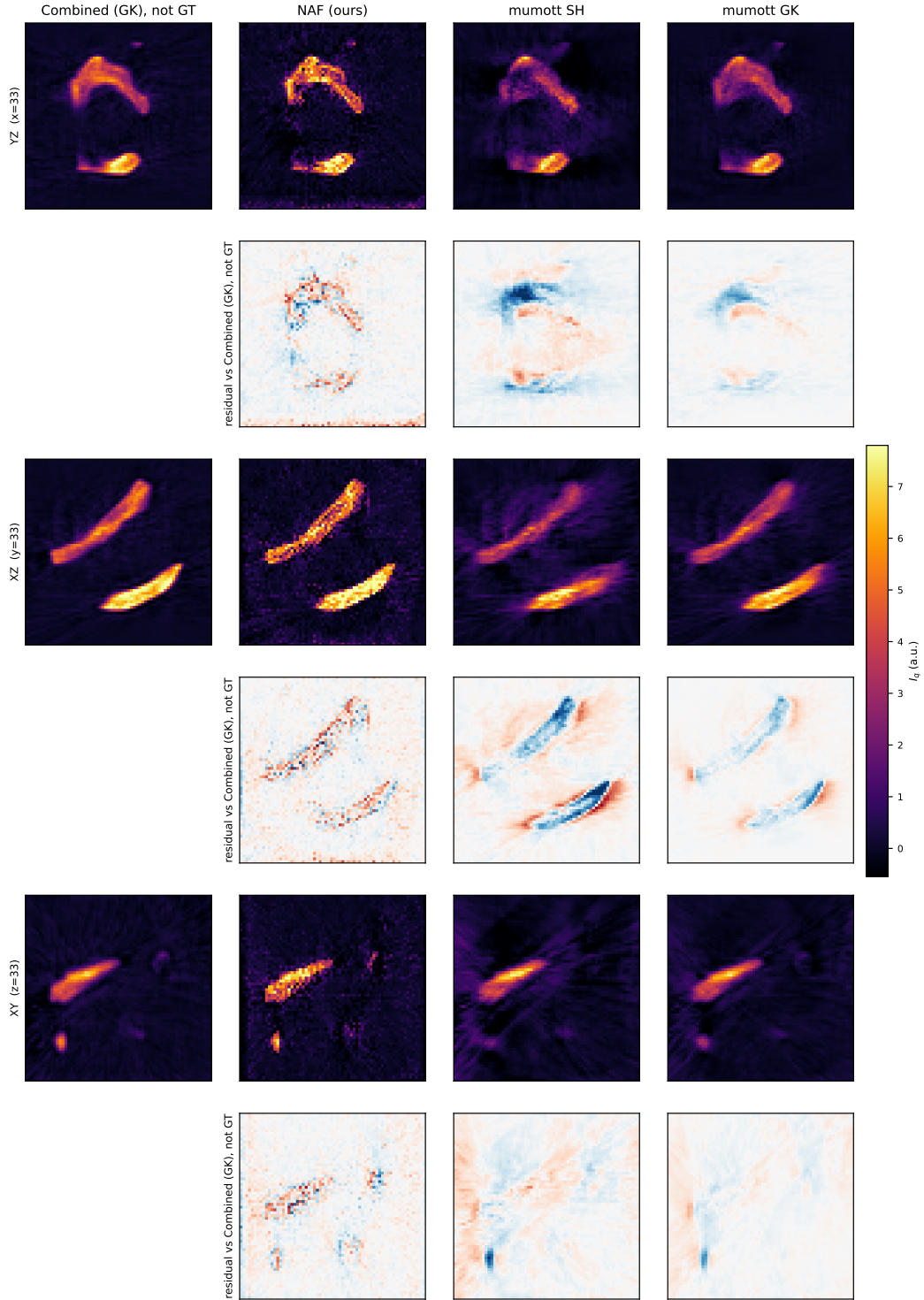


Figure 4: The same bone dataset along a fully measured direction (missing arc 0°). With complete angular coverage the methods largely agree, isolating the missing wedge as the source of the differences above.